# Exercise 4: Worked solutions

Electrical Engineering I · Module 92315 · Exercise 4

From Fundamentals to AI Apocalypse · personal study guide

## Contents


**Task 9: Field at a flat dielectric boundary** **2**

**Task 10: Two-dielectric coaxial capacitor** **3**

**Answers** **4**


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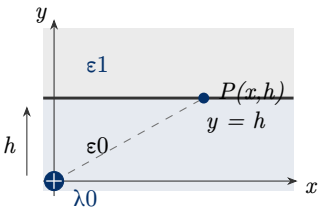
Try each part before opening the answer. We apply the interface conditions to the measured field, then use cylindrical Gauss surfaces and voltage integration for the layered coaxial geometry.

## Task 9: Field at a flat dielectric boundary

An infinite line charge of density $\lambda_0$ lies along the z-axis. The lower half-space y < h is air with permittivity $\epsilon_0$, and the upper half-space y ≥ h has permittivity $\epsilon_1$. In the air, measurements give the field below.

$$E_0(\mathrm{x},\mathrm{y}) = \lambda_0/[\pi(\varepsilon_0 + \varepsilon_1)(\mathrm{x}^2 + \mathrm{y}^2)]\,[\mathrm{x}e_\mathrm{x} + (\varepsilon_1/\varepsilon_0)\mathrm{y}e_\mathrm{y}]$$

**Connected topic:** Material boundary conditions (§1.11) keep tangential E continuous and normal D continuous when no free surface charge is present.




**Resolve the measured field into interface-parallel and interface-normal parts.** The x-component stays continuous; the y-component changes with permittivity.

### 9: Electric field in the dielectric at y = h

Try first: evaluate the measured air field at the interface, then apply the tangential-E and normal-D boundary conditions component by component.

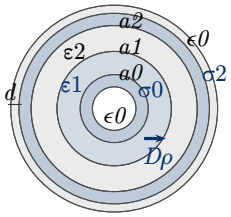
*Answers: see "Task 9: Field at a flat dielectric boundary" under Answers at the end.*

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## Task 10: Two-dielectric coaxial capacitor

Two infinitely long concentric metal cylinders share the z-axis. The inner metal's outer radius is $a_0$; the outer metal's inner radius is $a_2 > a_0$. Between them, permittivity is $\epsilon_1$ for $a_0 < \rho < a_1$ and $\epsilon_2$ for $a_1 < \rho < a_2$. The conductor walls each have thickness $d$; outside the conductors is material $\epsilon_0$. The only charges are $\sigma_0$ at the inner conductor's outer surface and $\sigma_2$ at the outer conductor's inner surface.

**Connected topics:** Gauss's law (§1.10) finds the radial flux; capacitance (§1.13) follows from charge per length divided by voltage.




**Track each cylindrical boundary by radius.** The field runs radially through $\varepsilon_1$ then $\varepsilon_2$; free surface charges sit on the two metal faces bordering the dielectric gap.

In the dielectric gap, symmetry gives $D = D_\rho(\rho)\,e_\rho$. The source specifies the voltage convention by $U_{20} = \int$ from $P_0$ (inner) to $P_2$ (outer) of $E \cdot ds$; use that orientation below.

### 10a: Flux density, field, and outer surface charge

Try first: apply cylindrical Gauss surfaces in each dielectric, then use continuity of normal D at $\rho = a_1$ and the conductor boundary at $\rho = a_2$.

*Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.*

### 10b: Field and $\sigma_0$ as functions of $U_{20}$

Try first: integrate E radially from $a_0$ to $a_2$, splitting at $a_1$, then solve the voltage relation for $\sigma_0$.

*Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.*

### 10c: Capacitance for length $\Delta z$

Try first: use the inner conductor's charge over length $\Delta z$ and divide by the voltage from 10b.

*Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.*

### 10d: Field inside the inner conductor and outside the outer one

Try first: use Gauss's law and the conductor condition $D = 0$, remembering that the two facing surface charges cancel per unit length.

*Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.*

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# Answers

## Task 9: Field at a flat dielectric boundary

### 9 · Apply both interface conditions

1. The interface is horizontal, so its normal is along y and its tangent is along x. There is no free charge sheet in the problem. Therefore:

$$E_{1\mathrm{x}} = E_{0\mathrm{x}}; \quad D_{1\mathrm{y}} = D_{0\mathrm{y}}, \quad \text{equivalently } \varepsilon_1 E_{1\mathrm{y}} = \varepsilon_0 E_{0\mathrm{y}}$$

2. First evaluate the given air field at y = h. Its tangential component is:

$$E_{0\mathrm{x}}(\mathrm{x}, \mathrm{h}) = \lambda_0 \mathrm{x}/[\pi(\varepsilon_0 + \varepsilon_1)(\mathrm{x}^2 + \mathrm{h}^2)]$$

3. Keep that x-component unchanged across the boundary:

$$E_{1\mathrm{x}}(\mathrm{x}, \mathrm{h}) = E_{0\mathrm{x}}(\mathrm{x}, \mathrm{h})$$

4. The measured normal component contains the factor $\varepsilon_1/\varepsilon_0$. Multiplying it by $\varepsilon_0/\varepsilon_1$ to preserve D cancels that factor:

$$E_{1\mathrm{y}}(\mathrm{x}, \mathrm{h}) = (\varepsilon_0/\varepsilon_1) E_{0\mathrm{y}}(\mathrm{x}, \mathrm{h}) = \lambda_0 \mathrm{h}/[\pi(\varepsilon_0 + \varepsilon_1)(\mathrm{x}^2 + \mathrm{h}^2)]$$

5. **Result:** combine the two components into the dielectric-side vector:

$$E_1(\mathrm{x}, \mathrm{h}) = \lambda_0/[\pi(\varepsilon_0 + \varepsilon_1)(\mathrm{x}^2 + \mathrm{h}^2)]\,[\mathrm{x}e_{\mathrm{x}} + \mathrm{h}e_{\mathrm{y}}]$$

**Check:** the tangential component is unchanged; the normal flux density matches on both sides. With SI inputs, E is in V/m (equivalently N/C).

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## Task 10: Two-dielectric coaxial capacitor

### 10a · Use enclosed charge per unit length

1. Choose a coaxial Gaussian cylinder of radius $\rho$ and length $\Delta z$, with $a_0 < \rho < a_2$. Radial D is constant over its curved side; its end caps contribute no flux. The enclosed free charge is the inner surface charge on length $\Delta z$:

$$D_\rho(\rho)\, 2\pi\rho\, \Delta z = \sigma_0\, 2\pi a_0\, \Delta z$$

2. Cancel the common factors. No free charge lies at the dielectric boundary $\rho = a_1$, so normal D has the same value in both layers:

$$D(\rho) = \sigma_0 a_0/\rho\, e_\rho, \qquad a_0 < \rho < a_2$$

3. Use $D = \varepsilon E$ separately in each material:

$$E(\rho) = [\sigma_0 a_0/(\varepsilon_1 \rho)]e_\rho, \qquad a_0 < \rho < a_1$$

$$E(\rho) = [\sigma_0 a_0/(\varepsilon_2 \rho)]e_\rho, \qquad a_1 < \rho < a_2$$

4. At the inner face of the outer metal cylinder, the normal from the conductor into the gap points inward, $-e_\rho$. Thus $\sigma_2 = D \cdot (-e_\rho)$ at $\rho = a_2$:

$$\sigma_2 = -\sigma_0 a_0/a_2$$

**Result:** D is radial and continuous through the dielectric interface; E changes magnitude inversely with permittivity. The two metal-face charges have opposite signs.

**Check:** $\sigma_2\ 2\pi a_2 \Delta z = -\sigma_0\ 2\pi a_0 \Delta z$, so the two facing surfaces carry equal and opposite charge.

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## Task 10: Two-dielectric coaxial capacitor

### 10b · Integrate through both layers

1. The specified path moves outward, in the $+e_\rho$ direction. Insert the two E expressions from 10a into the source's voltage integral:

$$U_{20} = \int_{a0}^{a1} [\sigma_0 a_0/(\varepsilon_1 \rho)]\, d\rho + \int_{a1}^{a2} [\sigma_0 a_0/(\varepsilon_2 \rho)]\, d\rho$$

2. Each integral of $1/\rho$ gives a logarithm:

$$U_{20} = \sigma_0 a_0[\ln(a_1/a_0)/\varepsilon_1 + \ln(a_2/a_1)/\varepsilon_2]$$

3. Define the positive geometry/material factor:

$$S = \ln(a_1/a_0)/\varepsilon_1 + \ln(a_2/a_1)/\varepsilon_2$$

4. Rearrange for the inner surface charge density, then substitute it back into D and E:

$$\sigma_0 = U_{20}/(a_0 S); \qquad D(\rho) = [U_{20}/(S\rho)]e_\rho$$

$$E(\rho) = [U_{20}/(\varepsilon_1 S\rho)]e_\rho, \qquad a_0 < \rho < a_1$$

$$E(\rho) = [U_{20}/(\varepsilon_2 S\rho)]e_\rho, \qquad a_1 < \rho < a_2$$

5. **Result:** these expressions use the path orientation specified in the sheet. Positive $U_{20}$ gives positive $\sigma_0$ and an outward field.

   **Check:** integrating the resulting E over both layers returns $U_{20}$. The field has units V/m and $\sigma_0$ has units C/m².

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## Task 10: Two-dielectric coaxial capacitor

### 10c · Capacitance per unit length

1. The magnitude of charge on the facing inner surface over length $\Delta z$ is $Q = \sigma_0\, 2\pi a_0 \Delta z$. Capacitance is charge divided by voltage:

$$C = Q/U_{20} = (\sigma_0 2\pi a_0 \Delta z)/U_{20}$$

2. Substitute $\sigma_0 = U_{20}/(a_0 S)$. The voltage and $a_0$ cancel:

$$C = 2\pi \Delta z/S; \quad C_B = C/\Delta z = 2\pi/S$$

3. **Result:** C is the capacitance over length $\Delta z$; $C_B$ is the capacitance loading per unit length.

   **Check:** S has units m/F, so C is in farads and $C_B$ in F/m. If $\varepsilon_1 = \varepsilon_2 = \varepsilon$, then $S = \ln(a_2/a_0)/\varepsilon$, recovering the homogeneous coaxial result.

## Task 10: Two-dielectric coaxial capacitor

### 10d · Zero field outside the charged gap

1. For $0 \le \rho < a_0$, a coaxial Gaussian cylinder encloses no free charge: $\sigma_0$ lies at $\rho = a_0$, outside this Gaussian surface. Thus its radial flux and field are zero. Within the inner metal wall D is also zero by the conductor condition.

2. For $\rho > a_2$, a Gaussian cylinder outside both facing surfaces encloses their total charge per length:

$$q_{enc}/\Delta z = 2\pi a_0 \sigma_0 + 2\pi a_2 \sigma_2 = 0$$

   The cancellation uses $\sigma_2 = -\sigma_0 a_0/a_2$. Therefore D and E vanish outside the outer conductor; they also vanish within its metal wall.

3. **Result:**

$$E = 0, \quad 0 \le \rho < a_0$$

$$E = 0, \quad \rho > a_2$$

   **Check:** the applied voltage and field are confined to the annular dielectric gap in this ideal infinitely long, concentric arrangement.

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